

## BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN ANIMALS

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (6 Lessons)*

| SUB-STRAND OVERVIEW             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Grade Level                     | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Subject                         | Biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Strand                          | Strand 3.0: Anatomy and Physiology of Animals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Sub-Strand                      | Sub-Strand 3.2: Growth and Development in Animals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Total Duration                  | 6 lessons × 40 minutes = 240 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Sub-Strand Content              | <ul style="list-style-type: none"> <li>– Metamorphosis (complete and incomplete metamorphosis in insects)</li> <li>– Role of growth hormones in insects</li> <li>– Growth curves in animals (continuous and intermittent growth)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Learning Outcomes               | <p>By the end of the sub-strand, the learner should be able to:</p> <ul style="list-style-type: none"> <li>a) distinguish between complete and incomplete metamorphosis in insects,</li> <li>b) describe the role of hormones in growth and development in insects,</li> <li>c) analyse growth curves in animals,</li> <li>d) appreciate the importance of growth and development in animals.</li> </ul>                                                                                                                                                                                                                                                                           |
| Core Competencies               | <ul style="list-style-type: none"> <li>– Digital Literacy: The learner appropriately selects animations illustrating the effect of hormones on insect growth and development (Lesson 5).</li> <li>– Creativity and Imagination: The learner adjusts and rethinks their ideas while modelling life cycles of different insects (Lessons 1 and 6).</li> <li>– Critical Thinking and Problem Solving: The learner analyses growth curves and calculates growth rates at different growth phases on sigmoid and intermittent curves (Lessons 2 and 3).</li> </ul>                                                                                                                      |
| Core Values                     | <ul style="list-style-type: none"> <li>– Love: The learner handles insects with compassion as they collect data on animals to calculate growth rates at different growth phases (Lesson 1).</li> <li>– Unity: The learner collaborates with others while plotting graphs to illustrate complete and incomplete metamorphosis in insects (Lessons 2 and 3).</li> </ul>                                                                                                                                                                                                                                                                                                              |
| Science & Engineering Practices | <ul style="list-style-type: none"> <li>– Developing and Using Models: Students build and revise life cycle models of grasshoppers (incomplete) and houseflies (complete) from Lesson 1 to Lesson 6.</li> <li>– Analyzing and Interpreting Data: Students plot and interpret intermittent and sigmoid growth curves and calculate growth rates (Lessons 2 and 3).</li> <li>– Obtaining, Evaluating, and Communicating Information: Students research insect hormones and evaluate animations and flowcharts (Lessons 4 and 5).</li> <li>– Engaging in Argument from Evidence: Students build Claim-Evidence-Reasoning posters answering the Driving Question (Lesson 6).</li> </ul> |

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| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Animal Welfare: The learner treats animals with care while collecting data for plotting growth curves in animals (Lessons 1 and 2).</li> <li>– Good Governance: The learner demonstrates principles of leadership while working with peers to search for information on insect hormones and their roles in growth and development (Lesson 4).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Agricultural Entomologists: study insect life stages to time interventions on pests such as locusts and armyworms in Kenyan farms.</li> <li>– Veterinary Researchers: use growth curves to monitor the health and development of livestock.</li> <li>– Parasitologists: study the life cycles and hormonal control of vectors such as mosquitoes and tsetse flies.</li> <li>– Wildlife Biologists: track growth and development of wild animals in places like the Rift Valley and Maasai Mara.</li> <li>– Horticultural Pest Control Specialists: design pest control that targets insect hormones in flower and vegetable farms around Naivasha.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Focus for Lessons</b>                          | This unit uses the puzzle of an empty insect skin beside a larger, newly emerged insect to drive inquiry into how animals grow. Students move from observing life cycles to plotting growth curves, then to the hormones that control moulting and metamorphosis. They keep a Driving Question Board and revise a cumulative model each lesson, using low-cost local specimens and printable data sheets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <p>1. How does growth in animals occur?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Anchoring Phenomenon</b>                       | On a maize leaf or tree bark in a school garden, students find a crunchy, empty, perfectly shaped insect skin (exuvia) with a split down its back, hanging beside a newly emerged grasshopper or cicada that is clearly larger than the skin it just left. Alongside it are photos or samples of caterpillars that become motionless pupae and later emerge as butterflies. This is puzzling: how can an animal grow bigger than a hard shell it was wearing, why does it not grow smoothly, and why do some insects change shape completely while others look like small adults? Students often think insects peel like sunburned skin or that caterpillars simply melt and rebuild without control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Storyline Thread</b>                           | <p>Lesson 1: Insects follow two distinct developmental pathways: complete metamorphosis (4 stages) vs. incomplete metamorphosis (3 stages).   Observe live specimens/specimen cards (grasshopper nymphs vs. housefly larvae/pupae); sort them into developmental sequences; construct initial comparative models.   Preserved insect specimens, hand lenses, life cycle diagram cards.   Why do grasshoppers increase in length in sudden jumps rather than smooth, daily growth?</p> <p>Lesson 2: Animals with exoskeletons exhibit intermittent (stepwise) growth curves because moulting (ecdysis) must occur before body dimensions increase.   Collect measurement data from insect nymph stages; plot length/mass vs. time to construct intermittent growth curves.   Measurement rulers, insect growth dataset worksheets, graph paper.   How do non-exoskeletal animals or organisms with internal skeletons grow compared to insects?</p> <p>Lesson 3: Vertebrates and soft-bodied animals exhibit continuous growth represented by a sigmoid (S-shaped) curve with distinct growth phases.   Plot mass/height data over time for mammals/birds; calculate growth rates across lag, exponential, deceleration, and stationary phases.   Animal growth rate data tables, graph paper, calculators.   What triggers and controls the precise timing of moulting and metamorphosis in insects?</p> <p>Lesson 4: Specific hormones (e.g., Ecdysone, Juvenile Hormone, Brain Hormone) interact to regulate moulting and stage transitions in insect life cycles.   Search information on insect hormones; analyze experimental flowcharts showing hormone gland removal/injection effects in larvae.   Reference texts, hormone function flowcharts, digital research devices.   Can visual animations</p> |

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|                             | <p>help us see how hormone concentration balances dictate larva-to-pupa shifts?</p> <p>Lesson 5: Fluctuations in Juvenile Hormone levels dictate whether an insect moults into another nymph/larva or transitions into a pupa/adult.   Select and view digital animations showing hormonal regulation in insect metamorphosis; synthesize findings into interactive flow diagrams.   Digital devices/projector, insect development animation links, flow chart templates.   How do all these growth patterns and chemical controls ensure animal survival and species continuity?</p> <p>Lesson 6: Controlled growth and developmental transitions allow animals to optimize resource use, reduce parent-offspring competition, and survive harsh environments.   Revisit and refine initial Day 1 models; construct Claim-Evidence-Reasoning (CER) posters answering the main Driving Question.   Chart paper, markers, DQB sticky notes, CER template.   Unit complete — How might human pesticides disrupt these exact hormonal pathways in agricultural pests?</p> |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– Locust nymphs in the Rift Valley and northern Kenya gradually gain wings across stages, while housefly maggots look nothing like adult flies.</li> <li>– Maize farmers see armyworm caterpillars feeding heavily and then vanishing into pupae in the soil before moths emerge.</li> <li>– A graph of a calf's or chick's mass over time forms a smooth S-shape, unlike the staircase pattern of a grasshopper's length.</li> <li>– Pesticides used on Kenyan farms can stop insect pests from moulting properly, which links to hormone control.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (80 minutes (double lesson)): The Empty Skin on the Maize Leaf: Two Ways Insects Grow

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Purpose</b>              | To launch the anchoring phenomenon (an empty insect skin beside a larger, newly emerged insect) and to have learners discover through evidence that insects follow two distinct developmental pathways: complete metamorphosis (4 stages) and incomplete metamorphosis (3 stages).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to distinguish between complete metamorphosis (egg, larva, pupa, adult) and incomplete metamorphosis (egg, nymph, adult) using specimens and life cycle cards (SLO a: distinguish between complete and incomplete metamorphosis in insects).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Skills</b>               | Predicting, observing and recording with a hand lens, sorting specimens into developmental sequences, constructing initial comparative models (NGSS SEP: Developing and Using Models), asking questions (NGSS SEP: Asking Questions), and stating a first Claim-Evidence-Reasoning (NGSS SEP: Engaging in Argument from Evidence).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Attitudes</b>            | Core competency: Creativity and Imagination - the learner adjusts and rethinks their ideas while modelling life cycles of different insects. Value: Love - the learner handles insects with compassion. Value: Unity - learners collaborate in sorting and modelling. PCI: Animal Welfare - the learner treats animals with care.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | This lesson opens the unit by presenting the puzzle of the empty exuvia and the larger insect beside it. Learners establish the two developmental pathways as a foundation, surface their ideas (such as 'insects peel like sunburn' or 'caterpillars melt and rebuild'), and open the DQB. The new question it raises for Lesson 2 is: Why do grasshoppers increase in length in sudden jumps rather than smooth, daily growth?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Safety Notes</b>         | Animal welfare: collect insects only from the school garden or compound by hand or soft net, take only the minimum needed (about 2 to 3 per group), never kill, pin or injure live insects for this lesson, and use preserved specimens only from already-dead or ethically sourced material. Keep live insects in ventilated containers (jars with perforated lids, a leaf and a damp cotton wool piece) in shade, and return them to where they were found within the same day. Handle gently with clean dry hands or a soft brush, and wash hands with soap afterwards. Do not handle insects if you have allergies; use hand lenses and avoid contact with eyes. Wear gloves for preserved specimens (preservative may irritate), keep jars closed, and avoid tasting or sniffing. Do not collect wasps, bees, hairy caterpillars or any insect that stings or irritates. Where internet is limited, use the printed data sheets and cards. |

### B. LESSON OVERVIEW

Learners begin in the school garden or at a display table with a real exuvia (a split, crunchy skin from a grasshopper or cicada on a maize leaf) beside a larger insect. They predict how the insect got bigger than its skin, draw a before-and-after diagram, observe preserved specimens and live grasshopper nymphs and housefly larvae/pupae with hand lenses, sort specimen and life cycle cards into sequences, and compare the two pathways in a model. A Gallery Walk lets peers review the models, and learners post questions on a new DQB. They leave with a first CER claim and the question: why do grasshoppers grow in sudden jumps rather than smoothly?

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                      | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES for readings</a><br>Search terms: growth development animals exuvia | Learners gather around a display (or the school garden maize plants) showing an empty exuvia with a split down its back next to a larger grasshopper nymph or adult, plus photos of a caterpillar, a motionless pupa and a butterfly emerging. Individually (3 min) they draw a two-stage before-and-after diagram of how the insect grew, annotating where they think the old shell went and what happened inside it. Then Think-Pair-Share: they compare drawings with a partner and write one prediction: 'I predict the insect got bigger than its skin because...'. Common ideas are recorded, such as 'it peels like sunburned skin' or 'the caterpillar melts and rebuilds'. | Hold up the exuvia and say: 'This was alive this morning, hanging on a maize leaf in our school garden. Look closely. The skin is smaller than the insect beside it. What do you notice? What do you wonder?' Give WAIT TIME of 10-15 seconds in silence before taking any answer. Say: 'Draw first, talk second. There are no wrong ideas today; I want to see your real thinking.' During pair talk, circulate and listen for the 'peeling like sunburn' and 'melting caterpillar' ideas without correcting. Cold-call 4 pairs (not volunteers) with: 'Tell us what your partner predicted, not what you predicted.' Record 4-5 ideas on the board using learners' own words. Say: 'Keep your drawing; we will revisit it at the end of the lesson.' Remind learners about gentle handling: 'These are living things; we handle them with compassion.' | Predict-Observe-Explain (POE) and Think-Pair-Share: Drawing a prediction before instruction makes learners commit to an idea, so later evidence creates a genuine 'need to know'. Pair sharing lets every learner rehearse their idea, and explaining a partner's idea builds listening and exposes the range of misconceptions that the class will test. | Observe and note on a class checklist: (1) Does the drawing show two stages with the old shell placed somewhere (e.g., 'dried up', 'peeled off', 'left behind')? (2) Does the learner show the insect inside the skin, or a skin that stretches or shrinks? (3) Does the learner use words like 'melt', 'peel' or 'stretch'? Tally the number of learners who hold the 'peeling' or 'melting' idea. Keep drawings in the learner portfolio as the baseline model.                                             |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES</i>                                                                                                                                                                                                       | In groups of 4 (assigned roles: handler, recorder, sorter, reporter) learners rotate through the evidence stations using hand lenses: Station 1: live grasshopper nymphs (or specimen cards) of different sizes plus an exuvia; Station 2: housefly larvae (maggot), pupae (puparia) and adults (preserved specimens or cards); Station 3: butterfly/caterpillar and pupa photo cards (e.g., cabbage white butterfly on sukuma wiki). At each station they sketch what they see, record features on a print-based data sheet (body shape, wings present or absent, legs, mobility,                                                                                                  | Before distribution say: 'Insects are living animals. Hands clean, movements slow, and every insect returns to the garden today.' Check that jars are in the shade and lids have holes. Say: 'Use your hand lens 2 cm from the specimen. Describe what you see, not what you think it should be.' Circulate with prompts: 'Does this young grasshopper look like the adult? What is different? What is the same?', 'Where are the wings on the maggot? On the pupa?', 'What do you see on the exuvia that you can also see on the live insect, such as eyes, leg joints or mouthparts?' After each prompt                                                                                                                                                                                                                                                | Predict-Observe-Explain (Observe step): Direct comparison of real specimens gives learners first-hand evidence about stage number and body shape. Sorting cards into sequences forces them to apply criteria (does the young look like the adult? is there a resting stage?), moving from description to pattern recognition.                             | Oral probes at each station: 'How do you know this is the same animal at a different stage?' Look for observable indicators: learner uses the hand lens correctly; the data sheet records at least 3 differing features between larva and adult; the grasshopper sequence has no pupa and the housefly sequence has a pupa; the learner states 'nymphs look like small adults' or 'larvae look nothing like adults'. Record on the checklist which groups sorted correctly first time and which need support. |

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| <p>library for this activity.</p> <p> <a href="#">Search ARES for readings</a></p> <p>Search terms: growth development animals exuvia</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>feeding, size), and note whether the young look like small adults. Then they sort the life cycle diagram cards into two developmental sequences: Sequence A (egg, nymph, nymph, adult) for grasshopper and Sequence B (egg, larva, pupa, adult) for housefly. They count the number of stages in each.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>give WAIT TIME of 10-15 seconds before moving on. Cold-call 3 groups to share one surprising observation. If a group sorts incorrectly, do not correct; ask: 'What evidence made you put this card here?' Ensure all insects are gently returned or kept in the shade, and collect preserved specimens afterward.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Organism growth and the environment</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p> <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b></p> <p><i>No closely matching reading in the ARES library for this activity.</i></p> <p> <a href="#">Search ARES for readings</a></p> <p>Search terms: growth development animals exuvia</p> | <p>Groups build an initial comparative model on a large sheet or manila paper: two columns (grasshopper: incomplete metamorphosis; housefly: complete metamorphosis) showing the stages in order with labelled drawings, and arrows showing where the exuvia (old skin) is left. They add a short caption: 'What changes? What stays the same?' Then they write a first Claim-Evidence-Reasoning (CER) statement: Claim: 'Insects follow two developmental pathways.' Evidence: observations from the specimens and cards (e.g., 'the housefly has a motionless pupa stage but the grasshopper does not'). Reasoning: why the evidence supports the claim. They then revisit their original Predict drawing and mark in a different colour anything they would now change. Finally, each group prepares to present.</p> | <p>Model one CER example on the board using a different Kenyan insect: 'Claim: A mosquito from Lake Victoria shores follows complete metamorphosis. Evidence: its young is a wriggler in water and it has a pupa. Reasoning: ...' Say: 'Notice I did not say "I think". I used evidence.' Give sentence frames: 'I observed...', 'This shows...', 'Therefore...'. Circulate and ask: 'Why do you think the grasshopper nymph has no pupa stage?', 'What does the arrow to the exuvia mean in your model?' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to state their CER aloud (not volunteers). Address the misconception gently: 'Some of you drew the old skin peeling like sunburn. Look at the exuvia again: what do you notice about the split on its back?' Do not give the full explanation; say: 'We have a puzzle: the insect is bigger than its skin. How? That is for our next lessons.'</p> | <p>Claim-Evidence-Reasoning (CER) modelling: Explicit modelling by the teacher shows learners how to link observations to a conclusion, building scientific argumentation. Building a comparative model makes thinking visible and gives a testable representation that will be revised in later lessons.</p> | <p>Check each group's model against a quick rubric: (1) Grasshopper shows 3 stages and no pupa; (2) Housefly shows 4 stages including pupa; (3) Exuvia is shown; (4) CER has a claim, at least one observation as evidence, and a reason. Oral probe: 'Which part of your evidence best supports your claim?' Note groups using 'I think' instead of evidence for follow-up.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Organism growth and the environment</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Learners display the phenomenon images and the driving question in the centre of a chart or wall space: 'How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?' Individually, they write 2 questions</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Say: 'A good question starts with how or why, and cannot be answered with yes or no.' Model: 'Who can turn "Does the skin break?" into a how or why question?' WAIT TIME 10-15 seconds. Cold-call 5 learners to place a question on the DQB and</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Gallery Walk and DQB: Peer review of models exposes learners to alternative ideas and strengthens critique skills. The DQB gives learners ownership of the inquiry and keeps the puzzle visible, so that each later lesson answers learner-generated</p>                                                   | <p>Use a peer rubric during the Gallery Walk with 3 criteria: stages in correct order, evidence shown, clear labels. Look for observable indicators: at least 80% of learners post two questions; questions are 'how/why' types; stars and wonders reference specific</p>                                                                                                        |

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| <p><i>Related topic: not an exact match for this lesson.</i><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p><b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms: growth development animals exuvia</p>                                                                                                              | <p>on sticky notes or slips of paper (one about the shell, one about the shape change). In groups they cluster questions into categories (e.g., How does the shell split? What controls growth? What happens inside the pupa? Why do some insects not have a pupa?). Then they take part in a Gallery Walk: groups walk around to view other groups' models and DQB clusters, leaving one 'star' (a strength) and one 'wonder' (a question) on sticky notes.</p>                                                                                                                                            | <p>explain where it fits. Highlight the question 'Why does the grasshopper get bigger suddenly rather than gradually?' if it arises, or introduce it yourself: 'I noticed that the smallest nymph and the next one are very different in size, but there is nothing in between. Why might that be?' During the Gallery Walk, give 6 minutes with a signal to rotate every 2 minutes. Say: 'Be kind, be specific, and use evidence in your feedback.' Circulate and note questions that link to later lessons (hormones, growth curves).</p>                                                                                                                                                                           | <p>questions.</p>                                                                                                                                                                                                                                                                                         | <p>features (e.g., 'You showed the pupa clearly'). Sort questions into themes to plan Lessons 2-6.</p>                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Model Building Phase</b></p> <p><b>VIDEO:</b><br/><a href="#">Organism growth and the environment</a><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p><b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms: growth development animals exuvia</p> | <p>Learners return to their original Predict drawing and, using a different colour, revise it to show what they now think about the insect and its old skin, noting what they are still unsure of (e.g., 'How did the insect get out? Why is the new insect bigger?'). Then they complete an exit ticket: (1) Name the stages for the grasshopper and the housefly. (2) Which pathway has a pupa? (3) One question you still have about how the insect got bigger than its skin. They add the first entry to the class Summary Table (What we observed / What we learned / What we can explain so far).</p> | <p>Say: 'Take out your first drawing. With a different coloured pen, change anything you now think is different. It is fine to keep ideas that you still believe.' Allow 3 minutes of silent work. Cold-call 3 learners to share one revision: 'What did you change and what evidence made you change it?' WAIT TIME 10-15 seconds. Then pose the bridging question: 'We know the stages. But why does the grasshopper grow in sudden jumps instead of a little each day? We will test that next lesson.' Collect exit tickets at the door. Remind learners to return live insects to the garden gently and wash hands with soap. Say: 'Bring any empty insect skins you find at home or in the shamba tomorrow.'</p> | <p>Cumulative model revision with exit ticket: Revising the initial model in a different colour makes learning visible and shows learners how their thinking has changed with evidence. The bridging question links this lesson to the next (growth in sudden steps), keeping the storyline coherent.</p> | <p>Exit ticket check: (1) correct stage lists for both insects; (2) pupa linked to complete metamorphosis; (3) a genuine question about growth or the shell. Observable indicators for the revised drawing: uses a second colour, adds at least one change based on evidence, and identifies one remaining uncertainty. Sort tickets into 'secure', 'partly secure' and 'needs support' groups and plan short re-teaching at the start of Lesson 2.</p> |

## D. TEACHER REFLECTION

Which learners still held the 'peeling like sunburn' or 'caterpillar melts' ideas after the Observe phase, and what evidence did they cite? Did every group handle live insects with care and return them safely? Which questions on the DQB can anchor Lessons 2 to 4 (shell and moulting, hormones, growth curves)? Did learners without internet access use the printed data sheets and cards effectively? What would I change about group roles, wait time or the sequencing of stations next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                       |                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe?                   | We observed an empty insect skin (exuvia) with a split on its back beside a larger insect; grasshopper nymphs that look like small adults with no pupa; housefly larvae that look nothing like the adult and a motionless pupa stage.                 |
| What did I learn?                     | Insects follow two developmental pathways: incomplete metamorphosis (egg, nymph, adult: 3 stages) as in the grasshopper, and complete metamorphosis (egg, larva, pupa, adult: 4 stages) as in the housefly and butterfly.                             |
| How does this explain the phenomenon? | We can explain that the two pathways differ in the number of stages and in whether the young resemble the adult, but we cannot yet explain how the insect got bigger than its old skin or why grasshoppers grow in sudden jumps rather than smoothly. |

## LESSON 2 (80 minutes (double lesson)): Stepwise Growth: Why Hard-Shelled Animals Grow in Jumps

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners collect and plot length and mass data from insect nymph stages to construct intermittent (stepwise) growth curves, and explain that moulting (ecdysis) must occur before body dimensions can increase in animals with exoskeletons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Knowledge</b>            | a) distinguish between complete and incomplete metamorphosis (early foundation); c) analyse growth curves in animals (intermittent growth). Learners explain that a rigid exoskeleton limits size, so length increases only after ecdysis, giving a staircase-shaped curve with plateaus (intermoult periods) and sudden rises (moulting).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Skills</b>               | Collecting measurement data (length, mass) from datasets and, where possible, live specimens; plotting line graphs on graph paper; interpreting growth curves; constructing claims supported by evidence and reasoning (CER); NGSS practices: Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Constructing Explanations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Attitudes</b>            | Love: handle insects with compassion while collecting data. Unity: collaborate with peers while plotting and comparing graphs. Appreciation of the importance of growth and development in animals. Digital Literacy and Creativity and Imagination are supported as learners select and rethink their models. Animal Welfare (PCI): treat animals with care. Good governance (PCI): demonstrate leadership in group roles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Purpose in Storyline</b> | Lesson 1 gave learners the phenomenon: an empty exuvia beside a larger newly emerged grasshopper, and caterpillars becoming pupae. Learners wondered how an animal can outgrow a hard shell. Lesson 2 gives the first quantitative evidence: growth in nymphs is stepwise, not smooth. This explains why the animal is bigger than its shell (it swelled and expanded after splitting the old skin) and sets up Lesson 3 (growth rate calculations with standard formulas) and the new question: How do non-exoskeletal animals or organisms with internal skeletons grow compared to insects?                                                                                                                                                                                                                                                                                                            |
| <b>Safety Notes</b>         | Animal welfare first: collect only a few insects (e.g., 3-5 grasshoppers or nymphs per class) from the school garden using soft containers or hands, never crushing them. Provide ventilated containers with maize leaves and sukuma wiki leaves, keep them shaded, and never hold them longer than 10 minutes. Measure with a ruler by gently placing the insect in a clear container or measure exuviae/photos instead of handling. Release every insect at the exact collection site the same day. Do not collect protected, stinging or biting species (wasps, bees, driver ants); avoid venomous insects. Wash hands after handling. Where specimens are unavailable or internet is limited, use the printed dataset worksheets. Follow national animal welfare standards: no harm, no killing, minimal handling, and prompt release. Learners with allergies should record data rather than handle. |

### B. LESSON OVERVIEW

Learners return to the school garden phenomenon: a grasshopper emerges larger than the hard exuvia it left on a maize leaf. They first predict what a graph of a nymph's length over time would look like (smooth line or steps?). They then measure real or dataset-based nymph lengths and masses (such as a locust or grasshopper nymph from Kisumu or Rift Valley farms) with rulers, plot them on graph paper, and observe plateaus and sudden jumps. Using CER, they explain that the exoskeleton cannot stretch, so the animal grows only right after moulting while the new cuticle is soft. A gallery walk allows peer review. The Driving Question Board is updated with new questions about internal skeleton animals, and the class revises its model of how an insect gets bigger than its shell.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sensemaking Strategy                                                                                                                                                                                                                                                                                  | Formative Assessment Strategy                                                                                                                                                                                                                                                                                         |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Growth models: introduction</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Linear vs. exponential growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners view the exuvia and emerged grasshopper from the school garden (specimen or photo). Individually they sketch on graph paper axes what they think the graph of a nymph's body length against time looks like over 10 weeks (smooth curve, straight line, or steps), and write one sentence explaining why, linking to how the animal became bigger than its shell. They use Think-Pair-Share to compare predictions and record a pair prediction on a sticky note.                                                                                                                                                                                   | Hold up the exuvia: "Last lesson we found this empty skin next to a bigger grasshopper. If we measured this grasshopper every few days from hatching, what would its growth graph look like?" Give WAIT TIME of 10-15 seconds in silence before anyone writes. Allow 2 minutes for individual sketching, 3 minutes for pairing. Then cold-call 4 pairs, ensuring at least 2 different shapes are shared. Ask: "What in your idea of the shell made you draw it that way?" Record all predictions on the board without judging. Remind learners of animal welfare rules for later handling.   | Predict-Observe-Explain (POE) and Think-Pair-Share: committing to a prediction first exposes the common misconception that growth is smooth, like a child's height, creating cognitive conflict when real data shows steps. Pair talk lets every learner voice reasoning before the class discussion. | Observe whether sketches show smooth lines versus steps; tally on a checklist how many learners link the shell to growth limits. Look for: (1) a labelled graph with axes, (2) a written reason. Use the tally to decide how much scaffolding to give during plotting.                                                |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Growth models: introduction</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Linear vs. exponential growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i>                                                                                                                                                                                     | In groups of 4 with defined roles (measurer, recorder, plotter, welfare officer), learners collect data. Where live nymphs from the school garden are available, they gently place them in a clear container over a ruler, and measure length (mm) and, if a balance is available, mass (g) for each instar; otherwise they work with the printed insect growth dataset worksheet (e.g., desert locust nymph lengths over 35 days with moulting dates in Rift Valley/Turkana conditions). They plot length against time on graph paper, join points, and mark the days when moulting occurred. They then describe the shape: plateaus and sudden rises. They | Model one data point on the board: "Day 10, length 12 mm. Where does it go on the axes?" Circulate and ask oral probes: "What is happening to the length between Day 12 and Day 18?" and "Why might the length not change even though the insect keeps eating sukuma wiki?" Give WAIT TIME of 10-15 seconds after each probe. Cold-call 3 groups to report the day of a moult. Check the welfare officer is enforcing gentle handling and a 10-minute limit. Provide print-based sheets for groups lacking specimens. Do not give the explanation yet; prompt: "Mark where the graph jumps." | Predict-Observe-Explain (observe step): learners generate the pattern themselves from data, so the staircase shape is evidence they discovered, not a fact they were told. Graphing converts numbers into a visual pattern that can be compared with their prediction.                                | Growth curve graph check: use a checklist at each group: correct axes labels with units, even scale, all points plotted accurately, moulting days marked, line shows plateaus and rises. Oral probes during modelling record whether learners can point to a plateau and a jump. Note groups needing help with scale. |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>also plot mass for comparison and note that mass may increase more gradually between moults while length jumps.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Growth models: introduction</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Linear vs. exponential growth: from data</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners use Claim-Evidence-Reasoning (CER) to answer: Why does the nymph's length increase in steps? The teacher models one CER on the board; groups then write their own on a CER template. Claim: growth in animals with exoskeletons is intermittent. Evidence: specific data from their graph (e.g., length stayed at 14 mm from Day 12-17, then jumped to 19 mm on Day 18). Reasoning: the hard, non-stretching exoskeleton limits size, so the animal must moult (ecdysis), split the old cuticle (leaving the exuvia), and swell while the new cuticle is soft before it hardens. They then do a Gallery Walk, posting graphs and CERs, and use a peer rubric to leave two stars and one wish on each.</p> | <p>Model: "My claim is... my evidence is Day 18... my reasoning connects to the exuvia on the maize leaf." Ask: "How does this explain why the grasshopper is bigger than its shell?" WAIT TIME 10-15 seconds. Cold-call 3 learners to give reasoning aloud, then 2 to challenge or add. Address the misconception that insects peel like sunburned skin: "Is the exuvia a thin peel, or a complete hard casing with legs and eyes covers?" During the Gallery Walk (8 minutes, rotating every 2 minutes), ask: "Which graph gives the strongest evidence, and why?" Briefly introduce that hormones (ecdysone) trigger moulting, to be explored later, and that growth rate calculations come in Lesson 3.</p> | <p>Claim-Evidence-Reasoning (CER) modelling and Gallery Walks: CER forces learners to tie their claims to specific data and a scientific mechanism, while peer review exposes them to alternative graphs and strengthens argumentation.</p> | <p>Peer rubric during the gallery walk: criteria are claim stated clearly, evidence quotes data with numbers and units, reasoning links exoskeleton and ecdysis. Check that at least 80% of CERs mention the non-stretching exoskeleton and moulting before growth. Exit ticket: "Draw a stepwise curve and label where moulting occurs; explain in one sentence why the insect is larger than its exuvia."</p> |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Growth models: introduction</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Linear vs. exponential</a></p>                                                                                                                                                                     | <p>Learners revisit the class DQB with the driving question about outgrowing a hard shell and why some animals change shape while others just get bigger. On sticky notes they add (a) what they now know (e.g., growth is stepwise, moulting comes first) and (b) new questions, such as: How do animals with internal bones like goats, cattle or humans in Kenya grow? Is their growth also in steps? What triggers moulting? Why do butterflies have a pupa stage? They place notes under categories such as Answered, Partly Answered and New.</p>                                                                                                                                                               | <p>Say: "Look at our board. Which questions did today's graphs help us answer, and which new ones did they raise?" Give WAIT TIME of 10-15 seconds before sharing. Cold-call 4 learners to place and justify a note. Highlight the planned question: "How do non-exoskeletal animals or organisms with internal skeletons grow compared to insects?" and invite learners to predict for next lesson. Move answered notes to the answered column with evidence from the graphs.</p>                                                                                                                                                                                                                              | <p>Driving Question Board with Think-Pair-Share: learners see their learning progress and generate the next investigation questions, building ownership and a coherent storyline from lesson to lesson.</p>                                 | <p>Review the quality of new questions: look for those that are testable and linked to today's evidence. Count how many learners contribute a question and how many link to the phenomenon. Use the notes to plan Lesson 3.</p>                                                                                                                                                                                 |

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| <a href="#">growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                  |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Growth models: introduction</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Linear vs. exponential growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a> | In their groups, learners revise their model from Lesson 1 by drawing a labelled diagram sequence of a grasshopper nymph: the nymph inside a tight exoskeleton, the old cuticle splitting along the back, the insect expanding with a soft new cuticle, and the hardening of the new cuticle. Beside it they paste a small version of their stepwise graph, linking each plateau and jump to a stage in the diagram. They add one sentence on why the emerged grasshopper is larger than the exuvia. They list what they still cannot explain (the shape change in butterflies; hormone control). Groups present a 1-minute oral explanation, and peers use rubrics. | Say: "Show me where on your model the growth actually happens, and where the graph shows it." Circulate and ask oral probes: "What would happen to the nymph if it could not moult?" WAIT TIME 10-15 seconds. Cold-call 3 groups to present, then ask 2 other learners to restate the key idea. Encourage creativity and rethinking of ideas. Close with the exit ticket and hand-out of the print dataset for Lesson 3 growth-rate calculations. Remind learners to release any held insects at the original site and wash hands. | Cumulative model building with peer review: learners integrate graph evidence with the mechanism of ecdysis into a visual model, which makes their thinking visible and revisable. Peer rubrics encourage collaborative critique (Unity). | Peer rubric during presentations: model includes (1) exoskeleton labelled, (2) moulting and exuvia, (3) expansion before hardening, (4) link to graph steps. Oral probes record whether learners can explain why growth occurs only after moulting. Exit ticket collected and sorted into ready, nearly ready and needs support. |

## D. TEACHER REFLECTION

Which learners still believed growth in insects is smooth or that the exuvia is just peeled skin after the graph work? Did every group handle insects gently and release them? Were the print datasets sufficient where specimens were unavailable? Which CER components (evidence with numbers, reasoning about the exoskeleton) were weakest, and how will you address them before Lesson 3 growth-rate calculations? What new questions on the DQB can seed the comparison with animals with internal skeletons?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We measured and plotted the length and mass of insect nymphs over time and saw that the graph rose in steps: flat periods with no length change, followed by sudden jumps when the insect moulted, leaving an empty exuvia.                                 |
| <b>What did I learn?</b>                     | Animals with a hard exoskeleton cannot grow smoothly because the shell does not stretch. They must moult (ecdysis) first, then swell while the new cuticle is soft, giving an intermittent (stepwise) growth curve.                                         |
| <b>How does this explain the phenomenon?</b> | The grasshopper on the maize leaf is bigger than its exuvia because it split the old shell, expanded its body before the new cuticle hardened, and so grew in a sudden jump. New question: how do animals with internal skeletons grow compared to insects? |

## LESSON 3 (80 minutes (double lesson)): Growth Curves: Why Do Vertebrates and Soft-Bodied Animals Grow Smoothly in an S-Shape While Insects with Hard Shells Grow in Steps?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Students gather quantitative evidence that vertebrates and soft-bodied animals grow continuously, giving an S-shaped (sigmoid) curve with four phases, and contrast this with the stepwise growth of the moulting insect in our phenomenon.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) analyse growth curves in animals (continuous and intermittent); (b) describe the lag, exponential (log), deceleration and stationary phases of a sigmoid curve; (c) calculate growth rate using $\text{Growth rate} = (\text{final value} - \text{initial value}) / \text{time interval}$ .                                                                                                                                                                                                                                                          |
| <b>Skills</b>               | NGSS SEP: Analysing and Interpreting Data (plot mass/height against time, identify trends); Using Mathematics and Computational Thinking (calculate growth rates with standard formulas); Constructing Explanations (CER). Learners collect, plot and interpret height, length or mass data and calculate growth rates at different growth phases.                                                                                                                                                                                                                                                                |
| <b>Attitudes</b>            | Core competencies: Digital Literacy (selecting suitable data sources or animations when available) and Creativity and Imagination (adjusting and rethinking ideas when revising models). Values: Love (handling any live animals with compassion while collecting data) and Unity (collaborating while plotting graphs). PCIs: Animal Welfare (treating animals with care when collecting data for growth curves) and Good Governance (demonstrating leadership roles in groups). Appreciate the importance of growth and development in animals.                                                                 |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | In Lessons 1-2 learners saw the empty exuvia beside a larger grasshopper and learned that a hard exoskeleton forces growth in steps (intermittent growth). Lesson 3 adds the contrast case: animals without a rigid outer shell, such as Kenyan chicks, calves, goats and humans, grow continuously in a sigmoid curve. Learners calculate growth rates to show that growth is not random but has phases. This sets up the new question for Lesson 4: What triggers and controls the precise timing of moulting and metamorphosis in insects?                                                                     |
| <b>Safety Notes</b>         | Mostly desk-based data analysis, so low risk. If any live insects (for example grasshoppers from the school garden) are shown as a comparison, collect only a few using hands or soft containers, handle gently, keep in ventilated containers with maize leaves, do not harm or squeeze them, avoid hot sun, wash hands afterwards, and release them at the exact place of collection at the end of the lesson in line with national animal welfare standards. Do not collect from protected areas. Calculators and graph paper only; no sharp tools needed. Data sheets are printed so no internet is required. |

### B. LESSON OVERVIEW

Learners return to the exuvia phenomenon and ask: if the grasshopper grows in sudden jumps because of its hard shell, how do animals without such a shell, such as a Kenyan broiler chick from Kiambu, a Friesian calf from Nyahururu or a human child, grow? Learners first predict the shape of a mass-against-time graph. They then plot printed data tables (chick mass over 10 weeks, calf mass over 2 years, a child's height over 20 years, and a comparison grasshopper nymph length with stepped moult data), calculate growth rates for each phase using  $\text{Growth rate} = (\text{final} - \text{initial}) / \text{time}$ , and identify lag, exponential, deceleration and stationary phases. They use CER to explain why the vertebrate curve is smooth while the insect curve is a staircase, add to the Driving Question Board, and revise their class model. The lesson ends with the question: what controls when an insect moults?

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                     | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Exponential and logistic growth in populations</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Linear vs. exponential growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings | (10 min) Learners look at the phenomenon photo again: the exuvia beside the bigger grasshopper. The teacher asks: 'The grasshopper grew in jumps. What would a graph of a Kenyan broiler chick's mass over 10 weeks look like?' Individually, learners sketch a prediction on a blank axis (mass on y, weeks on x) and write one sentence: 'I think the graph will look like ____ because ____.' They then Think-Pair-Share and compare sketches with a partner, noting whether the line is straight, stepped, or curved. Learners also predict: Does a chick have an exuvia like the grasshopper?                                                                             | Display the exuvia photo and say: 'Last lesson we saw the grasshopper grow in steps because of its hard shell. Today we test whether every animal grows like that.' Pose the question, then give WAIT TIME of 10-15 seconds of silent thinking before sketching. Say: 'Draw what you honestly think; there are no wrong predictions today.' Circulate and note three or four different sketch types (straight line, staircase, curve). During pair talk say: 'Tell your partner why you chose that shape.' Cold-call 4 pairs with different sketches to draw theirs on the board, asking 'What reasoning led to that shape?' without evaluating. Keep all predictions visible to revisit. | Think-Pair-Share combined with the Predict step of Predict-Observe-Explain (POE). Committing to a prediction before seeing data exposes preconceptions (for example that all animals grow in a straight line or like insects). It creates cognitive conflict and a reason to examine the data later.                                     | Observable indicators: each learner has a sketch with labelled axes and a one-sentence reason. Tally on the board how many predicted straight line, staircase, or S-curve. Look for the misconception 'growth is a constant straight-line increase' or 'all animals grow in steps'. No grades; use findings to adjust how much support is needed in the Observe phase.                                                                           |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Per capita population growth and exponential growth</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Exponential vs. linear growth</a><br><i>Related topic: not</i>                                                                                                                                                                                                                          | (25 min) In groups of 4 with assigned roles (Data Recorder, Plotter, Calculator, Spokesperson - practising leadership and Unity), learners receive printed data sheets: Data Set A: Broiler chick mass (g) at weeks 0-10 from a Kiambu poultry farm; Data Set B: Child height (cm) from age 0-20 in Kenya; Data Set C: Grasshopper nymph body length (mm) over 60 days with moult days marked. Each group plots Set A on graph paper (mass against time), joins points with a smooth line, and labels where growth is slow, fastest, slowing and levelling. Using calculators and the formula $\text{Growth rate} = (\text{final value} - \text{initial value}) / \text{time}$ | Model the first calculation on the board: 'From week 3 at 300 g to week 4 at 520 g: $(520 - 300) / 1 = 220$ g per week.' Say: 'Always write the units.' Circulate with a checklist and ask oral probes: 'Which part of your graph is steepest? What does steepness tell you about the growth rate?' and 'Why is the growth rate near zero in the last weeks?' Give WAIT TIME of 10-15 seconds after each probe. Check graph accuracy: axes labelled, uniform scale, points plotted correctly. Say: 'Check your partner's plot before you draw the line.' Ask each group: 'What happened at the moult days in Set C?' Remind groups with live                                              | Predict-Observe-Explain (Observe step) with data analysis: learners build the evidence themselves by plotting and calculating rather than being told the shape. Turning numbers into a graph and rates makes the four phases visible and quantifiable, and comparing Set A with Set C shows how a rigid exoskeleton changes the pattern. | Growth curve graph check: use a quick checklist per group (axes labelled with units, even scales, points accurate within half a square, smooth S-shaped line, four phases labelled). Check the growth rate table: correct formula use and units (g/week). Oral probes: learners should say that exponential-phase growth rate is greatest and stationary-phase is near zero. Note groups who cannot connect steepness with rate for re-teaching. |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES</a> for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>value) / time interval, they calculate growth rate for four intervals: lag (e.g., weeks 0-1), exponential (e.g., weeks 3-6), deceleration (weeks 7-8) and stationary (weeks 9-10). They record answers with units (g/week) in a table. Faster groups plot Set B or C and compare. Learners note that the grasshopper graph stays flat between moults and jumps at moulting, unlike the smooth chick curve. If live grasshoppers are shown, they remain in ventilated containers and are handled gently.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>insects to keep them shaded and release them correctly afterward. Provide print-based sheets for all groups and extra calculators for sharing. Support struggling groups with a partly completed table.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Linear vs. exponential growth: from data</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b></p> <p><a href="#">Linear vs. exponential growth: from data</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES</a> for similar readings</p> | <p>(20 min) Gallery Walk: each group posts its graph and growth rate table on the wall. Groups rotate every 3 minutes with sticky notes giving one 'star' (what is correct) and one 'wish' (what could improve) using a peer rubric (accuracy of plot, correct calculation, clear phase labels). Back at their seats, each learner writes a CER response to the question: 'Why does the chick's graph form a smooth S-shape while the grasshopper's graph forms steps?' Claim: a one-sentence answer. Evidence: at least two data values or growth rates from the graphs. Reasoning: link to biology (the chick has no rigid exoskeleton so body cells and tissues can increase continuously; early growth is slow while the animal adjusts and has few cells, then cell division gives rapid growth, then resources and genetic limits slow growth until adult size is reached; the grasshopper's hard exoskeleton cannot stretch so size increases only after moulting). Learners then compare their weeks 0-10 graph with their earlier predictions and</p> | <p>Before the walk say: 'Look for evidence, not just neatness.' Circulate during the Gallery Walk and listen for correct vocabulary. Model a CER on the board with another example: 'Claim: growth slows after week 8. Evidence: rate fell from 220 to 40 g/week. Reasoning: the chick is nearing its adult size set by genes and food intake.' Then say: 'Now write yours.' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to read their reasoning and ask 'Who can add to that?' Probe: 'Why is the lag phase slow if the chick is young and about to grow?' Address misconceptions directly: 'Does the insect melt and rebuild? What does our data say about the steps?' Link to the phenomenon: 'So the grasshopper's exuvia is the cost of having a hard, non-stretching shell.' Collect the CER responses for review.</p> | <p>Claim-Evidence-Reasoning (CER) modelling together with Gallery Walks for peer review. Gallery Walks let learners see multiple representations and correct each other's errors, while CER forces them to justify the pattern with data and biological mechanism, linking the sigmoid curve to cell division, resource limits and the exoskeleton.</p> | <p>Peer rubric (stars and wishes) completed for at least two other groups. CER responses: scoring indicators - claim addresses smooth versus stepped growth; evidence includes at least two numerical values with units; reasoning mentions cell division, limiting factors or adult size, and the exoskeleton. Cold-call responses show whether learners can name the four phases in order. Exit ticket (given in the final 5 minutes): 1) Name the phase with the highest growth rate. 2) Calculate the growth rate if a calf rose from 40 kg to 70 kg in 10 weeks. 3) Write one question about what makes an insect start to moult.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | explain any change in thinking in pairs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Linear vs. exponential growth: from data (example 2)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Exponential vs. linear growth</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>(10 min) Learners revisit the Driving Question Board with the central question: 'How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?' Each learner writes one new question on a sticky note from today's work (for example: 'What tells the grasshopper nymph that it is time to moult?', 'Why do chicks stop growing at a certain size?', 'Do caterpillars show a sigmoid curve?', 'What controls the sudden jumps in the insect graph?'). They post it under the headings Growth Pattern, Control of Growth, and Metamorphosis. The class then moves earlier questions that today's evidence answered (for example, 'Do all animals grow the same way?') to an 'Answered' section, with the evidence cited.</p> | <p>Say: 'Take 2 minutes to write a question that today's graphs made you wonder.' Give WAIT TIME of 10-15 seconds before writing starts. Cold-call 4 learners to read and place questions, asking 'Which category does this fit, and why?' Move answered questions and ask 'What evidence from today lets us move it?' Highlight and read aloud the questions about what triggers and controls timing: 'Many of you ask what makes the insect moult at that moment. That is exactly our next investigation.' Keep the board visible in the classroom.</p>                            | <p>Driving Question Board with Think-Pair-Share: pairs briefly share their questions before posting. Posting and categorising questions shows learners that evidence generates new questions and gives them ownership of the learning path into Lesson 4.</p>                                                                          | <p>Observable indicators: every learner posts at least one question; at least some questions move from 'what' to 'what causes/controls' (higher-level thinking). Note the number of questions on hormonal control as a sign of readiness for Lesson 4. Listen to whether the reasoning given for moving answered questions cites data.</p>                                                                                                                                       |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Exponential and logistic growth in populations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Linear vs.</a></p>                                                                                                                                                                                                                                                                                                | <p>(15 min) In their groups, learners revise the class model of animal growth begun in Lessons 1-2. On a large sheet they add a new section: two side-by-side graphs (a smooth S-curve labelled with lag, exponential, deceleration, stationary phases and a staircase curve for the moulting insect) plus arrows and labels explaining why they differ (no rigid shell versus exoskeleton, cell division, limiting factors). They add the formula for growth rate with one worked example from their data. They use a different colour for new ideas and mark with a question mark the</p>                                                                                                                                                                                                   | <p>Say: 'Use a different colour for what we learned today so we can see how our thinking is changing.' Circulate and ask: 'Where on your model does the exuvia appear? What does the flat part of the insect graph represent?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to explain one change they made and why. Prompt for gaps: 'What in your model explains the timing of the steps? What is still missing?' Affirm that the question marks are scientific honesty. Close by saying: 'We now know how the growth patterns differ, but not what</p> | <p>Cumulative model revision with peer feedback (Gallery Walk style group presentations). Revising a visible model forces learners to integrate new quantitative evidence with earlier ideas, and marking unknowns turns gaps into the next investigation, supporting Creativity and Imagination as they adjust and rethink ideas.</p> | <p>Peer rubrics during presentations: model includes both curve types, four phases labelled correctly, growth rate formula with worked example, explanation connecting shell/no shell to the pattern, and at least one unanswered question. Oral probes during group modelling: learners can explain why the insect graph is flat between moults. Collect the exit tickets; a correct calculation of 3 kg per week and identification of the exponential phase show mastery.</p> |

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| <a href="#">exponential growth: from data</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES</a> for similar readings | part they cannot yet explain: what triggers and controls the timing of moulting and metamorphosis. Groups quickly present for 1 minute to another group, who give feedback using the peer rubric. | controls them. In Lesson 4 we investigate what triggers and controls the precise timing of moulting and metamorphosis in insects.' Remind learners that any live insects must be released gently where they were collected. |  |  |
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D. TEACHER REFLECTION

Which learners could connect steepness of the graph to growth rate, and who still treated the curve as a picture? Did the predictions show the misconception that all animals grow like insects or in a straight line? Were the print-based data sets and calculators enough for every group to finish four growth rate calculations, and which interval was hardest? Did the comparison between the chick and the grasshopper clarify the effect of the exoskeleton? How many DQB questions asked about control or triggers, and is the class ready for hormones in Lesson 4? If live insects were used, were they handled kindly and released safely at the point of collection?

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| What did I observe?                                          | We plotted mass and height data for a Kenyan broiler chick, a child, and a grasshopper nymph. The chick and child graphs formed smooth S-shaped curves, while the grasshopper's length stayed flat and jumped at each moult.                                                                                                                                                                                          |
| What did I learn?                                            | Vertebrates and soft-bodied animals grow continuously in a sigmoid curve with lag, exponential, deceleration and stationary phases. Growth rate = (final value - initial value) / time interval, and it is highest in the exponential phase and near zero in the stationary phase. Hard-shelled insects show intermittent, stepwise growth because the exoskeleton cannot stretch.                                    |
| How does this explain the phenomenon?                        | The chick has no rigid shell, so cells can divide and increase size continuously: slow at first, then rapid, then slowing as it nears adult size set by genes and resources. The grasshopper must shed its exoskeleton (leaving the exuvia) before it can grow, so its size increases in steps. What triggers and controls the timing of moulting and metamorphosis is still unanswered and is the focus of Lesson 4. |

## LESSON 4 (80 minutes (double lesson)): Hormones as the Control Switches of Moulting and Metamorphosis

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Purpose</b>              | To help learners figure out that the hormones Ecdysone, Juvenile Hormone and Brain Hormone interact to regulate moulting and stage transitions in insect life cycles, and so explain how a grasshopper or cicada can leave its hard exuvia behind and why some insects change shape completely while others only grow bigger.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Knowledge</b>            | Learners: (b) describe the role of hormones in growth and development in insects. They will explain that Brain Hormone from neurosecretory cells signals the prothoracic glands to release Ecdysone, which triggers moulting, and that Juvenile Hormone from the corpora allata decides what kind of moult occurs (larva to larva when JH is high, larva to pupa when JH is low, pupa to adult when JH is absent).                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Skills</b>               | Searching for information on insect hormones from reference texts and digital devices; analysing experimental flowcharts showing gland removal and hormone injection in larvae; Developing and using models (SEP); Analysing and interpreting data (SEP); Constructing explanations using Claim-Evidence-Reasoning (SEP). Core competency: Digital Literacy, where learners appropriately select reliable sources on insect hormones. Print-based data sheets are provided where connectivity is limited.                                                                                                                                                                                                                                                                                                        |
| <b>Attitudes</b>            | Love: handle insects and specimens with compassion. Unity: collaborate with peers when analysing flowcharts. PCI Good governance: demonstrate principles of leadership (rotating roles of reader, recorder, reporter, timekeeper) while working with peers to search for information on insect hormones and their roles in growth and development. PCI Animal Welfare: treat animals with care.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | Lessons 1-3 established that insects grow in steps (intermittent growth shown by growth curves and growth rates calculated in Lesson 3), and that some insects undergo complete metamorphosis while others undergo incomplete metamorphosis. Lesson 4 answers the 'what controls it?' question: hormones interact like a control system. This lesson gives the mechanism behind the exuvia beside the newly emerged grasshopper or cicada. It raises the new question for Lesson 5: Can visual animations help us see how hormone concentration balances dictate larva-to-pupa shifts?                                                                                                                                                                                                                           |
| <b>Safety Notes</b>         | Only observe live insects already collected for earlier lessons, or use preserved exuvia, from the school garden. Follow national animal welfare standards: collect only a few insects by hand or with a soft net, never kill, injure, freeze or pin insects for this lesson, keep them in ventilated containers with leaves of their food plant (maize or sukuma wiki), keep them in shade, wash hands before and after handling, and release them at the point of capture within 24 hours. Gland removal and hormone injection are studied ONLY through flowcharts and data sheets; no learner performs surgery or injections on any animal. Do not handle insects if a learner has an allergy; assign that learner to the reader or recorder role. Check that device use stays on approved educational sites. |

### B. LESSON OVERVIEW

Learners return to the anchoring phenomenon: the empty split exuvia beside a larger newly emerged grasshopper or cicada, and a caterpillar that becomes a motionless pupa. They predict what inside the insect could 'tell' it when to moult and when to change shape. In groups they search reference texts, hormone function flowcharts and digital sources (or print data sheets) to identify Brain Hormone, Ecdysone and Juvenile Hormone, then analyse experimental flowcharts of gland removal and hormone injection in larvae (for example: removing the head/brain region, tying off the body, removing the corpora allata, injecting extra JH). They use Claim-Evidence-Reasoning (CER) to explain how the hormones interact, add the mechanism to the class model and the Driving Question Board, and leave with a new question about how hormone concentration balances change over time, which leads to animations in Lesson 5.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sensemaking Strategy                                                                                                                                                                                                                                                    | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br>No closely matching video in the ARES library for this activity.<br><a href="#">Search ARES for videos</a><br>Search terms: growth development animals ecdysone<br><br> <b>READING:</b><br>No closely matching reading in the ARES library for this activity.<br><a href="#">Search ARES for readings</a><br>Search terms: growth development animals ecdysone | Learners look at the exuvia and the newly emerged grasshopper or cicada (specimen or photo) and a photo of a maize-leaf armyworm or other caterpillar becoming a pupa. Individually they write for 2 minutes: 'What might be inside the insect's body that tells it WHEN to moult and WHETHER to change into a completely different shape?' They then use Think-Pair-Share, and in groups predict what would happen to a caterpillar if the part that sends the 'moult' signal were removed. Predictions are recorded on a POE sheet (Predict column) with a sketch. | Display the exuvia and say: 'Last lesson we saw that growth happens in steps. Today we ask: who or what is the boss that switches these steps on?' Pose the question and give WAIT TIME of 10-15 seconds in silence before anyone writes. Give 2 minutes of silent writing, then 2 minutes pair talk. Cold-call 4 pairs (not volunteers): 'Pair 3, what did your partner predict and why?' Probe with: 'What makes you think so?' and 'Where in your own body do signals that control growth, like at puberty, come from?' Record 4-5 predictions on the board without judging them. Note misconceptions such as 'the shell just peels like sunburned skin' or 'caterpillars melt with no control'. Ask: 'If the signal were removed, what do you predict happens to the caterpillar?' Wait 10-15 seconds again. | Predict-Observe-Explain (POE) and Think-Pair-Share. Predicting first makes prior ideas visible and creates a need to know; Think-Pair-Share gives every learner rehearsal time so the cold-calls are fair and the class sees a range of ideas to test against evidence. | Observable indicators: every learner has a written prediction and a sketch on the POE sheet; at least 4 different predictions are shared; teacher tallies on the board how many predict a chemical or hormone-type signal versus 'the insect decides', temperature only, or 'the shell breaks by itself'. Look for learners linking to human puberty hormones as a sign of transfer from Form 1-3 ideas.                                                                            |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br>No closely matching video in the ARES library for this activity.<br><a href="#">Search ARES for videos</a><br>Search terms: growth development animals ecdysone<br><br> <b>READING:</b><br>No closely matching reading in the ARES                                                                                                                         | Part A (15 min): In groups of 4 with rotating leadership roles (reader, recorder, reporter, timekeeper), learners search reference texts, hormone function flowcharts and digital research devices (or print data sheets) to complete a 3-row table: Brain Hormone (PTTH), Ecdysone, Juvenile Hormone, with columns for where made, what it does, and when it is high or low. Part B (20 min): Groups analyse four experimental flowcharts on larvae (for example Experiment 1: larva with brain/neurosecretory cells removed, leading to no moult;                  | Circulate with a checklist. Before starting say: 'Choose reliable sources: a textbook, a teacher-approved site, or the print sheet. Note the source name.' Remind about animal welfare: 'The live insects stay in their ventilated containers in the shade; we study the experiments on paper, not on living animals.' Give each group 2 minutes to read each flowchart silently, then WAIT TIME of 10-15 seconds before group discussion. Ask probes: 'What was the variable the scientist changed?' 'What was the control?' 'What does                                                                                                                                                                                                                                                                         | Predict-Observe-Explain (Observe) with data analysis of flowcharts. Comparing 'gland removed' with 'hormone replaced' is the logic of a controlled experiment, so learners can infer cause and effect rather than just memorize hormone names.                          | Observable indicators: the hormone table is complete with correct sources noted; learners correctly identify the variable changed and the result in at least 3 of the 4 flowcharts; oral probes during group work show learners say 'when X was removed, Y stopped, so X is needed for Y'; leadership roles rotate (check the group role sheet); teacher checks for the common error of swapping the functions of Ecdysone (triggers moulting) and JH (maintains the larval state). |

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| <p><i>library for this activity.</i></p> <p> <a href="#">Search ARES for readings</a><br/>Search terms: growth development animals ecdysone</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Experiment 2: same larva injected with Ecdysone, leading to moult resumes; Experiment 3: young larva with extra Juvenile Hormone injected, leading to moults into a larger larva, not a pupa; Experiment 4: larva with corpora allata removed early, leading to a small, early pupa). They write what was changed, what happened, and what it shows. Relevant Kenyan context: the maize stalk borer or armyworm caterpillars on farms in the Rift Valley and the desert locust or grasshoppers (incomplete metamorphosis, nymphs) that moult several times. Learners then compare with the school-garden exuvia and discuss why the grasshopper nymph shows fewer shape changes (JH stays present until the final moult to adult).</p>                                         | <p>this result show about which gland sends the first signal?' Help a group that is stuck with: 'Follow the arrows: which gland acts first, which second?' Pause the class after Experiment 2 and cold-call 3 groups: 'What do you now say about the role of Ecdysone?' Support learners with limited connectivity by giving print sheets and pairing devices. After Experiment 4, cold-call 3 groups: 'Why does low JH change the type of moult?'</p>                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/><i>No closely matching video in the ARES library for this activity.</i></p> <p> <a href="#">Search ARES for videos</a><br/>Search terms: growth development animals ecdysone</p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i></p> <p> <a href="#">Search ARES for readings</a><br/>Search terms: growth development animals ecdysone</p> | <p>Individually then in pairs, learners write a Claim-Evidence-Reasoning (CER) statement answering: 'How does the grasshopper or cicada outgrow its hard exuvia, and what decides whether an insect becomes another larva or a pupa?' Teacher models a CER on a different example first (for example, why a locust nymph stays a nymph through 4-5 moults). Pairs then draw a flow diagram linking: Brain Hormone, prothoracic gland, Ecdysone, moult, and Juvenile Hormone high or low, giving larva-to-larva, larva-to-pupa, pupa-to-adult. Groups do a short Gallery Walk: charts are posted, learners leave sticky-note feedback ('one strength, one question') using a peer rubric on claim accuracy, use of the experimental evidence, and reasoning that uses the term</p> | <p>Model a CER aloud: 'My claim is... My evidence from Experiment 2 is... My reasoning is that...'. Then say: 'Now you write your own. You have 5 minutes.' After writing give WAIT TIME of 10-15 seconds before pair sharing. Cold-call 4 learners to read a claim aloud, asking 'Who can add evidence from a flowchart?' Challenge: 'Does your explanation say why the old skin has to be shed in a hard-shelled insect?' (The cuticle does not stretch; Ecdysone triggers a new cuticle to form and the old one to split, and the insect swells with air or water before the new one hardens.) Direct the Gallery Walk: 6 minutes, rotate every 2 minutes, with each learner placing at least 2 sticky notes. Debrief by cold-calling 3 groups: 'What was the strongest feedback you received?' Correct the</p> | <p>Claim-Evidence-Reasoning (CER) modelling and Gallery Walks. CER forces learners to link each claim to specific experimental evidence and a scientific reason, and the Gallery Walk uses peer review to expose gaps, so learners refine explanations before they commit to the class model.</p> | <p>Observable indicators using the peer rubric (0-2 for each of claim, evidence, reasoning): a strong response states that Brain Hormone stimulates the prothoracic gland to release Ecdysone which triggers moulting, and that high JH keeps the larval form while low or absent JH allows pupation and adult emergence; cites at least one experimental result; and uses terms correctly. Teacher notes which groups reverse the roles of JH and Ecdysone for reteaching. Oral probes while groups present: 'What would happen if JH stayed high at the last larval moult?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 'hormone'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | misconception that insects 'melt and rebuild without control': say 'There is a hormone control system, and the signals decide each step.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><i>No closely matching video in the ARES library for this activity.</i><br><a href="#">Search ARES for videos</a><br>Search terms: growth development animals ecdysone<br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br><a href="#">Search ARES for readings</a><br>Search terms: growth development animals ecdysone | Learners return to the Driving Question Board with the question 'How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?'. In groups they write on sticky notes (1) one question now answered by today's evidence (move to the 'Answered' area with a note of the evidence), and (2) one new question. The expected new question is about how the changing amounts of the hormones over time cause the shift from larva to pupa, which leads to Lesson 5. Learners also sort earlier questions into 'answered', 'partly answered' and 'still to investigate'. | Say: 'Look at our board. Which questions can we now answer using today's flowcharts?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to move a question and justify it with evidence from a specific experiment. Prompt for new questions: 'What do you still not understand about HOW MUCH of each hormone is present and WHEN?' Guide to the storyline question: 'Can visual animations help us see how hormone concentration balances dictate larva-to-pupa shifts?' Add it to the board in a different colour. Ask at least 2 learners with quieter voices (cold-call) to read out new questions. | Driving Question Board tracking. Moving questions between zones shows learners their own growing understanding and keeps the investigation learner-driven, while new questions create a purpose for Lesson 5.                         | Observable indicators: at least 3 earlier questions are correctly moved with a stated piece of evidence; each group contributes one new question that is testable or researchable; a question about hormone concentration or timing appears; exit ticket at the end (one sentence: 'Brain Hormone, Ecdysone and Juvenile Hormone work together by... ') is collected and sorted as secure, partial, or misconception.                                                                                                             |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><i>No closely matching video in the ARES library for this activity.</i><br><a href="#">Search ARES for videos</a><br>Search terms: growth development animals ecdysone<br><br> <b>READING:</b><br><i>No closely</i>                                                                                                                                                                       | Learners revise the class and individual models of insect growth by adding the hormone control layer to their earlier model of intermittent growth (the stepped growth curve from Lesson 3) and life cycles. Using the grasshopper (incomplete metamorphosis: egg, nymph, adult) and a moth or butterfly such as the maize armyworm (complete: egg, larva, pupa, adult), they draw arrows showing where Brain Hormone, Ecdysone and JH act at each moult, and annotate the stepped growth curve to show that each                                                                                                                  | Say: 'Take out your model from Lesson 3. What is missing that explains who controls the steps?' Give WAIT TIME of 10-15 seconds, then 8 minutes for revising. Circulate and ask during oral probes: 'Where on your growth curve does Ecdysone rise?' 'Why does the grasshopper nymph look like a small adult but the caterpillar does not?' Remind learners to mark what they changed in a different colour so revisions are visible. Cold-call 3 learners to share one change and the evidence that led them to it.                                                                                              | Cumulative model revision with a peer check. Adding the mechanism to a model learners already own shows that explanations are built and refined from evidence, and marking revisions in colour makes the change in thinking explicit. | Observable indicators: the revised model shows the correct order Brain Hormone, then prothoracic gland, then Ecdysone, then moult; shows JH high for larva-to-larva moults and low or absent for pupation and adult emergence; links moults to steps on the growth curve (growth curve graph check); distinguishes complete from incomplete metamorphosis using hormone action; colour-coded revisions are visible. Peer rubric during short presentations to a neighbouring group: accurate, evidence-linked, clear. Exit ticket |

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| <i>matching reading in the ARES library for this activity.</i><br><a href="#">Search ARES for readings</a><br>Search terms: growth development animals ecdysone | upward step coincides with moulting (Ecdysone peaks) and the plateau with the hardened cuticle. They write a short caption: 'The exuvia is left behind because...'. Creativity and Imagination: learners adjust and rethink their ideas as they revise. | Close by handing out the exit ticket, and point out the unanswered question: 'We now know WHICH hormones, but not HOW their amounts change over time. Next lesson we will watch animations to find out.' Remind learners to return all insects to the garden at the point of capture and wash hands. | collected and sorted. |
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## D. TEACHER REFLECTION

After the lesson ask yourself: Did learners use the experimental flowcharts as evidence rather than reciting hormone names? Which groups swapped the roles of Ecdysone and Juvenile Hormone, and what reteaching is needed at the start of Lesson 5? Were cold-calls spread fairly, including learners without devices who used print sheets? Did the waiting time of 10-15 seconds produce deeper answers? Were all insects handled gently and returned to the garden, in line with national animal welfare standards? Did the math of growth rate from Lesson 3 appear correctly on the growth curve annotations? Which misconceptions (shell 'peels like sunburned skin', caterpillars 'melt' without control) persist, and what evidence will address them in the animations next lesson?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>What did I observe?</b>                   | We analysed flowcharts of experiments on larvae: with the brain/neurosecretory cells removed the larva did not moult; with Ecdysone injected it moulted again; with extra Juvenile Hormone it moulted into another larva instead of a pupa; with the corpora allata removed early it formed a small pupa early. We also compared these with the empty exuvia and the larger grasshopper or cicada from the school garden.                                                               |
| <b>What did I learn?</b>                     | Brain Hormone signals the prothoracic glands to release Ecdysone, which triggers moulting. Juvenile Hormone decides the kind of moult: high levels keep the larval form, low levels allow pupation, and none allows adult emergence. Hormones interact like a control system, so moulting and shape change are regulated and not random.                                                                                                                                                |
| <b>How does this explain the phenomenon?</b> | The grasshopper or cicada outgrows its hard shell because hormones trigger a new, larger cuticle to form beneath the old one, the old one splits and is left as the exuvia, and the insect expands before the new cuticle hardens, so growth happens in steps. Insects with complete metamorphosis, like caterpillars, change shape completely because Juvenile Hormone falls and allows pupation, while grasshopper nymphs keep JH until the last moult and so look like small adults. |

## LESSON 5 (80 minutes (double lesson)): The Chemical Switch: How Juvenile Hormone Decides Whether an Insect Stays Young or Becomes an Adult

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Purpose</b>              | To help learners use digital animations to explain how fluctuating Juvenile Hormone (JH) levels, working with the moulting hormone ecdysone, decide whether an insect moults into another nymph/larva or transitions into a pupa/adult, and so explain the empty exuvia and the grasshopper-versus-butterfly puzzle from the school garden.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Knowledge</b>            | By the end of the lesson the learner should be able to: (b) describe the role of hormones in growth and development in insects; state that ecdysone triggers moulting while the level of JH determines the type of moult (high JH = nymph/larva to nymph/larva; falling JH = pupa then adult); contrast hormonal control in incomplete metamorphosis (grasshopper) and complete metamorphosis (butterfly/moth).                                                                                                                                                                                                                                                                                             |
| <b>Skills</b>               | Digital Literacy: the learner appropriately selects animations illustrating the effect of hormones on insect growth and development. Creativity and Imagination: the learner adjusts and rethinks their ideas while building flow diagrams. NGSS SEP: Developing and Using Models; Obtaining, Evaluating and Communicating Information; Constructing Explanations (CER).                                                                                                                                                                                                                                                                                                                                    |
| <b>Attitudes</b>            | Love: the learner handles insects and living specimens with compassion. Unity: the learner collaborates with others while building and reviewing flow diagrams. Good governance (PCI): the learner demonstrates leadership principles by taking rotating roles (selector, recorder, presenter, timekeeper) while searching for information on insect hormones. Animal Welfare (PCI): the learner treats animals with care.                                                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | Lessons 1-4 established that insects grow in stages, that the exoskeleton restricts growth so growth is intermittent (step-like) in the growth curve, and that moulting occurs. Growth rates were calculated in Lesson 3 using the standard formula: $\text{growth rate} = \frac{\text{change in measurement}}{\text{change in time}}$ . This lesson supplies the CAUSE: a chemical control system. It explains why the same moulting process produces a bigger nymph in a grasshopper but a pupa in a butterfly. It raises the next question: How do all these growth patterns and chemical controls ensure animal survival and species continuity?                                                        |
| <b>Safety Notes</b>         | Animal welfare: if live insects are shown (e.g., grasshoppers or caterpillars collected from the school garden or maize field), collect only a few by hand or soft container, handle gently, keep in ventilated containers with fresh leaves (sukuma wiki or maize leaves), never pin, crush or leave in the sun, and return them to the exact place of collection within the lesson/day. Wash hands after handling. Do not collect protected species, and do not handle stinging or hairy caterpillars with bare hands (use a leaf or paper to move them). Projector cables and devices should be kept away from walkways. Where internet is limited, use pre-downloaded animations or print-based sheets. |

### B. LESSON OVERVIEW

Learners return to the empty exuvia beside the larger newly emerged grasshopper and the motionless pupa of a butterfly. They predict what 'switch' inside the insect decides whether it stays a nymph/larva or becomes an adult. They then select and view digital animations (pre-downloaded on the teacher's device or projected; print-based storyboard sheets as the offline alternative) showing hormonal regulation of insect metamorphosis, record evidence, and synthesise it in flow diagrams on templates. A gallery walk with peer rubrics and CER statements consolidates understanding. The DQB is updated and the class model of 'how an insect outgrows its shell' is revised to include hormones. The lesson ends with an exit ticket and the question for Lesson 6 on survival and species continuity.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                               | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>HTML:</b><br><a href="#">Hormones and Plant Growth - Advanced (Flexbook)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES</a> for similar readings | (approx. 12 min) Learners look at the phenomenon display: an exuvia from the school garden on a maize leaf, a photo of the newly emerged larger grasshopper, and photos of a caterpillar, a pupa and a butterfly (e.g., a African monarch/ common citrus swallowtail from Kakamega or Kisumu gardens). Individually they write in their notebooks a prediction: 'What inside the insect's body tells it to moult, and what decides whether the next stage is a bigger nymph/larva or a pupa/adult?' They sketch a quick 'prediction model' with arrows. They do Think-Pair-Share, then share with the class. | Display the exuvia/photos. Say: 'Remember the empty skin on the maize leaf. Grasshopper nymphs come out looking like small adults, but the caterpillar becomes a pupa. Something is controlling this. Write your prediction silently for 2 minutes.' Then: 'Turn to your partner and share for 1 minute each.' Use WAIT TIME of 10-15 seconds after asking: 'What could be the signal inside the body that tells the insect which type of moult to make?' Cold-call 4 pairs (not volunteers) including at least 2 who have not spoken in earlier lessons. Say: 'Thank you, I am not judging right or wrong; we will test these ideas with evidence.' Record 3-4 prediction ideas on the board (e.g., 'temperature', 'food', 'a chemical in the blood', 'age'). Address the misconception that caterpillars 'melt and rebuild without control' by asking: 'If it is without control, why does it always make a butterfly and not something else?' (WAIT 10 seconds). | Predict-Observe-Explain (POE) and Think-Pair-Share: committing to a prediction first creates a cognitive 'need to know' and makes learners notice where the animations confirm or challenge their thinking. Pairing lets learners rehearse and refine ideas safely before public sharing.                                                          | Observable indicators: (1) at least 80% of learners have a written prediction with a drawn arrow model before pair-sharing; (2) teacher listens in on 5 pairs and tallies who mentions 'hormone/chemical', 'brain', 'gland', 'food' or 'temperature'; (3) misconceptions noted ('insects peel like sunburnt skin', 'melt and rebuild') on a quick checklist to revisit at the exit ticket.                                          |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos                                                                                                                                                                                                                                                                                                                                                                                          | (approx. 25 min) In groups of 4 with rotating roles (selector, recorder, presenter, timekeeper), learners select and view at least two digital animations from the teacher's list (e.g., 'Insect moulting and ecdysone', 'Hormonal control of metamorphosis in butterfly', 'Incomplete metamorphosis in locust/grasshopper'). The selector justifies why an animation is suitable using a 3-point checklist: scientifically accurate, shows hormone levels, shows both                                                                                                                                       | Before viewing, say: 'You are evidence hunters. Your job is not to watch passively but to capture what hormone is released, from where, and what happens to JH level at each moult.' Show the animations on the projector once through for the whole class, pausing at 3 key points to ask: 'What just happened to the JH level?' Use WAIT TIME of 10-15 seconds, then cold-call 3 learners per pause. Circulate and ask oral probes: 'Which gland released                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Predict-Observe-Explain (Observe): learners collect structured evidence to test their predictions. The structured observation table directs attention to the variable that matters (JH level) and builds the pattern 'high JH = stay juvenile; low JH = become adult'. Linking JH curves to the step-like growth curve integrates earlier lessons. | Oral probes during group work: learner can correctly state at least one of 'ecdysone triggers moulting' and 'high JH keeps the juvenile form'. Growth curve graph check: teacher checks that groups correctly mark moulting points on the step-like curve and do not draw a smooth line. Observable indicators: tables completed with JH levels for all stages; groups justify animation choice using the checklist; roles rotated. |

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| <p> <b>HTML:</b><br/> <a href="#">Hormones and Plant Growth - Advanced (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>complete and incomplete metamorphosis. The recorder fills an observation table with columns: Stage; Hormone(s) present; Level of JH (high/falling/low/absent); Result of moult. Groups without connectivity use the print storyboard sheets with JH-level graphs showing a JH curve across instars of a grasshopper and a butterfly across larva-pupa-adult. Learners also plot or read the JH-level curve against the stepwise growth curve from Lesson 3 and note that ecdysone peaks coincide with each step upward in size. Live specimens (grasshopper nymph, caterpillar) or preserved exuviae are available at one station for comparison and gentle observation only.</p>                                                                                                                                                                                          | <p>ecdysone? What would happen if JH stayed high at the last larval moult?' 'Where does your table show a difference between grasshopper and butterfly?'<br/> Remind learners: 'Handle the live insects with a soft leaf, gently, and they go back to the maize plants at the end.' Support groups using print sheets by checking they read the graph axes correctly. Assign leadership rotation: 'Every 8 minutes swap roles so every member practises leading.'</p>                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Organism growth and the environment</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Hormones and Plant Growth - Advanced (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>(approx. 25 min) Groups synthesise their findings into an interactive flow diagram using the flow chart template (digitally on devices, or with arrow cards and paper strips that can be moved around). The diagram must include: brain signals; prothoracic gland releasing ecdysone; corpora allata releasing JH; and two pathways (A) high JH: nymph/larva to larger nymph/larva; (B) JH falls: larva to pupa to adult (complete metamorphosis), or last nymph to adult (incomplete). Learners then write a Claim-Evidence-Reasoning statement answering: 'How does the same moulting process lead to a bigger grasshopper nymph in one case and a pupa in another?' Example claim: 'The level of JH decides the type of moult.' They then use the diagram to explain the exuvia in the school garden and why the new grasshopper is larger than the discarded skin</p> | <p>Model one CER example on the board using a different phenomenon: 'Claim: A locust nymph in a swarm in Turkana moults into another nymph. Evidence: JH level is high. Reasoning: High JH maintains juvenile features, so ecdysone-triggered moulting produces another nymph.' Say: 'Now write yours for the butterfly pupa.' Give 5 minutes. Use WAIT TIME of 10-15 seconds before cold-calling 3 groups to read claims. Probe: 'What is your evidence? Which part of the animation shows it?' 'What would you expect if JH were artificially added to a final-instar caterpillar?' (Expect: extra larval stage or a giant larva.) Link to Kenyan context: 'Scientists at icipe in Nairobi study insect hormones to control pests like locusts and the fall armyworm on maize farms.' Run the gallery walk: 'Read, then write one star (what is accurate)</p> | <p>Claim-Evidence-Reasoning (CER) modelling: separates what the learner believes from the data and the scientific principle, so explanations are accountable to evidence. Gallery Walk: peer review exposes learners to alternative representations and makes them revise and defend their reasoning (Creativity and Imagination: adjusting and rethinking ideas). Flow diagram: shows cause-and-effect and sequence of the hormonal control.</p> | <p>Peer rubrics (4 criteria scored 1-3) during the gallery walk; teacher checks CER statements for: a claim naming JH level, evidence cited from an animation or graph, reasoning that links JH to type of moult. Observable indicators: diagram includes both pathways and correct labels of ecdysone and JH; learners can explain, without notes, why grasshoppers have no pupa stage; sticky notes give at least one specific comment. Success criterion: 75% of groups include ecdysone as the moult trigger and JH as the decider.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (body expands before the new cuticle hardens). Then they hold a Gallery Walk: groups post diagrams, rotate every 3 minutes, and leave peer feedback sticky notes using the peer rubric (accuracy, completeness, clarity, creativity).                                                                                                                                                                                                                                                                                                                                                               | and one wish (what can be improved) on a sticky note. Be kind and specific.' Promote Unity: 'Give feedback that helps your classmates improve.' After the walk, ask 4 learners cold-call to report one idea they changed because of another group's diagram. Correct any misconceptions: JH is not the hormone that causes moulting; ecdysone is, and JH decides the kind of moult.                                                                                                                                                                                                                                                 |                                                                                                                                                                                               |                                                                                                                                                                                                                                |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Hormones and Plant Growth - Advanced (Flexbook)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | (approx. 8 min) Learners revisit the class Driving Question Board. In pairs, they review existing question cards and mark those now answered (e.g., 'Why does the caterpillar stop eating and become still?', 'What controls moulting?'). They write 1-2 new questions on cards, such as 'How do these growth patterns and chemical controls help the species survive?', 'Can farmers use JH chemicals to control pests such as armyworm on maize?', 'Do vertebrates like humans also use hormones for growth?' They place the cards on the board under categories: Answered, Partly answered, New. | Say: 'Look at our board. Which questions can we now answer using the evidence from today? Come and move a card and explain why.' Cold-call 3 learners to move cards and justify using evidence. Give WAIT TIME of 10-15 seconds when asking: 'What new question has today's evidence made you curious about?' Use cold-call for 3 learners to read their new questions. Highlight the question for Lesson 6 and read it aloud: 'How do all these growth patterns and chemical controls ensure animal survival and species continuity?' Praise questions linked to Kenyan contexts (pest control, locust swarms, silk moth rearing). | DQB with Think-Pair-Share: learners see their own learning progress and generate authentic questions that drive the next investigation, which keeps the storyline coherent and student-owned. | Observable indicators: at least 5 cards moved with an evidence-based justification; at least 6 new, testable questions posted; teacher notes questions that reveal lingering misconceptions (e.g., 'melting without control'). |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | (approx. 10 min) Learners revise their class/individual model of 'How does an insect outgrow its hard shell and why do some change shape completely?' Using the earlier model (exoskeleton,                                                                                                                                                                                                                                                                                                                                                                                                         | Say: 'Open your model from Lesson 4. In a different colour, add what you learned about hormones today. Show the arrows and the JH levels.' Walk around with oral probes: 'Show me where JH is                                                                                                                                                                                                                                                                                                                                                                                                                                       | Cumulative model revision: learners integrate new evidence into a single explanatory model, make their thinking visible, and reflect on how their ideas have changed since the Predict Phase. | Exit tickets: success if the learner names ecdysone as moulting trigger, JH as the determinant of the type of moult, and explains the absence of a pupa in incomplete metamorphosis. Observable                                |

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| <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b></p> <p><a href="#">Hormones and Plant Growth - Advanced (Flexbook)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>moulting, stepwise growth) they add: brain, prothoracic gland, corpora allata, ecdysone, JH, and the high-JH/low-JH pathways. They label their model for both a grasshopper and a butterfly, and annotate with a short note on what changed from their prediction. Individually they complete an exit ticket: (1) Which hormone triggers moulting? (2) What does a falling JH level cause in a butterfly caterpillar? (3) Draw or write why the grasshopper has no pupa stage. (4) One question you still have.</p> | <p>high in your model.' 'Why does the exuvia get left behind?' Use WAIT TIME of 10-15 seconds, and cold-call 3 learners to explain one revision to the class. Collect exit tickets at the door. Remind learners to return live insects to the garden gently and wash their hands. Give a bridging statement: 'Next time we ask why this whole system matters for survival and the continuation of the species.'</p> | <p>Exit ticket provides individual accountability.</p> | <p>indicators: model includes at least 4 new labelled components and a correct pathway for both insect types; learners can explain one change from their original prediction. Teacher sorts exit tickets into three piles (secure, partial, misconception) to plan re-teaching at the start of Lesson 6.</p> |
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## D. TEACHER REFLECTION

After the lesson consider: Did learners select animations using the checklist rather than choosing the most entertaining one? Which groups confused ecdysone with JH, and what evidence in the animation helped correct this? Did the print-based sheets give equal access to the evidence for learners without devices? Were live insects handled with care and returned safely, in line with national animal welfare standards? How well did learners connect hormone levels to the stepwise growth curves and the growth-rate calculations from Lesson 3 (rate = change in measurement ÷ change in time)? Did all learners participate in leadership roles? Which exit ticket misconceptions need to be revisited before Lesson 6?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>What did I observe?</b>                   | We viewed animations and graphs showing ecdysone peaks before every moult, and JH levels that stay high during nymph/larval moults but fall before the pupa and adult stages. We also saw that grasshoppers go from nymph to adult without a pupa, while butterflies have a pupal stage. The exuvia in the garden is the old exoskeleton left after a moult.                                                                                           |
| <b>What did I learn?</b>                     | Moulting is triggered by ecdysone, and the level of Juvenile Hormone decides what type of moult occurs: high JH means the insect moults into another nymph/larva, while falling JH leads to a pupa (complete metamorphosis) or an adult (incomplete metamorphosis). Growth is stepwise because the exoskeleton must be shed before the insect can get bigger. Metamorphosis is a controlled, hormone-regulated process, not 'melting without control'. |
| <b>How does this explain the phenomenon?</b> | An insect outgrows its hard shell by releasing ecdysone, which causes it to shed its old exoskeleton and expand before the new one hardens. A grasshopper keeps a high-to-declining JH level through several nymph moults and becomes an adult without a pupa, while a butterfly caterpillar's JH falls dramatically before it forms a pupa and then an adult, which is why some animals change shape completely and others just grow bigger.          |



## LESSON 6 (80 minutes (double lesson)): Final Explanation: Controlled Growth, Metamorphosis and Survival in Kenyan Insects

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Purpose</b>              | To consolidate evidence from Lessons 1-5 into a final scientific explanation of how an insect outgrows its hard exoskeleton and why some insects change shape completely while others only grow bigger, and to compare the final model with the Day 1 model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Knowledge</b>            | By the end of the lesson the learner should be able to: a) distinguish between complete and incomplete metamorphosis in insects; b) describe the role of hormones (ecdysone and juvenile hormone) in growth and development in insects; c) analyse growth curves in animals (continuous and intermittent growth); d) appreciate the importance of growth and development in animals.                                                                                                                                                                                                                                                                                                                                            |
| <b>Skills</b>               | Constructing evidence-based explanations (CER); revising and comparing models; interpreting intermittent (step-like) and sigmoid growth curves and using the standard growth rate formula (change in measurement divided by time, as taught in Lesson 3); communicating scientific ideas through posters and gallery walks; giving and receiving peer feedback using a rubric.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Attitudes</b>            | Love: handles insects and specimens with compassion. Unity: collaborates with peers when building CER posters and giving peer review. Good governance (PCI): demonstrates leadership in group roles (facilitator, recorder, presenter, materials manager). Animal Welfare (PCI): treats animals with care. Core competencies: Creativity and Imagination (adjusts and rethinks ideas while revising models); Digital Literacy (selects appropriate animations or print data sheets when internet is limited).                                                                                                                                                                                                                   |
| <b>Key Inquiry Question</b> | How does growth in animals occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | This is the final lesson of the unit. Learners return to the anchoring phenomenon (an empty grasshopper or cicada exuvia with a split down the back beside a larger newly emerged insect, and caterpillars becoming pupae then butterflies) and use all evidence to answer the Driving Question. Their Final Explanation shows that controlled growth and developmental transitions allow animals to optimize resource use, reduce parent-offspring competition, and survive harsh environments. It closes with a new question for extension: How might human pesticides disrupt these exact hormonal pathways in agricultural pests?                                                                                           |
| <b>Safety Notes</b>         | Handle all specimens and exuviae gently; wash hands with soap and water after handling. Any live insects collected from the school garden (maize leaves, sukuma wiki plants, tree bark) must be collected without harm following national animal welfare standards: collect only the minimum needed, use ventilated containers with leaves and a little moisture, avoid crowding, never expose to direct sun or heat, do not pin, crush or kill for display, and return all live insects to the exact place of collection at the end of the lesson. Empty exuviae may be used freely. Learners with allergies to insect bites or pollen should wear gloves or observe from photos. Keep walkways clear during the gallery walk. |

### B. LESSON OVERVIEW

Learners begin by predicting how their Day 1 model would change after what they have learned, then observe their original Day 1 models side by side with evidence from Lessons 1-5 (exuvia samples, life-cycle models, hormone animations or print summaries, and growth curve graphs of grasshopper/nymph length against time showing step-like growth). They construct Claim-Evidence-Reasoning (CER) posters answering the Driving Question, take part in a Gallery Walk with peer rubrics, complete the Driving Question Board (DQB) by sorting sticky notes into answered and new questions, and revise their models into a final model with a Day 1 versus final comparison. An exit ticket checks individual understanding and the lesson ends with the new question about pesticides and insect hormones.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                           | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES</a> for readings<br>Search terms: growth development animals exuvia | Learners look at the anchoring phenomenon displayed on the board: a photo or real exuvia from a maize leaf beside a larger newly emerged grasshopper or cicada, plus a photo of a caterpillar, pupa and butterfly. Individually, in 3 minutes, they write a prediction on a sticky note: 'If I redrew my Day 1 model today, the three biggest things I would change are...' Then they Think-Pair-Share with a partner and predict which parts of the Driving Question they can now answer fully, partly or not yet.                                                                                                                   | Display the phenomenon and say: 'Remember the empty skin we found on the maize leaf? Look carefully at the skin and the insect next to it. Today you will explain how this insect outgrew its own hard shell, and why a butterfly looks nothing like its caterpillar while a grasshopper nymph looks like a small adult.' Give 3 minutes of silent writing. Then say: 'Turn to your partner and share your predictions for 2 minutes.' Circulate and listen. Cold-call 4 pairs: 'What would you change in your Day 1 model?' WAIT TIME: 10-15 seconds after each question before calling on anyone. Record common predictions on the board under 'What we think we know now'. | Think-Pair-Share combined with Predict-Observe-Explain (POE): predicting before revisiting evidence makes learners activate their ideas from Lessons 1-5 and exposes remaining misconceptions (for example, that insects peel like sunburned skin or that caterpillars melt and rebuild without control), so that the next phase of observation has a purpose. | Observable indicators: the sticky note names at least two specific changes (for example, adding hormones, showing the old exoskeleton splitting, or adding a pupal stage); learners use vocabulary such as moulting, ecdysone or nymph; during pairing, both partners speak. Teacher records on a tally sheet which learners still mention peeling or melting, to target in later phases.                                                                                                                                  |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Organism growth and the environment</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i>                                                                                                                                                                              | In groups of four with assigned roles (facilitator, recorder, materials manager, presenter), learners revisit their evidence folders: (1) their original Day 1 models; (2) exuviae or photos of grasshopper and cicada moults; (3) their life-cycle models of complete metamorphosis (for example butterfly, housefly or mosquito from Lake Victoria region) and incomplete metamorphosis (grasshopper, cockroach); (4) the hormone summary (ecdysone triggers moulting, juvenile hormone keeps the juvenile form) from animations or print-based data sheets; (5) growth curve graphs from Lesson 3 showing intermittent (step-like) | Say: 'Open your evidence folders. Your job is to be detectives: find three pieces of evidence that explain the empty skin.' Distribute print-based data sheets to groups without connectivity or tablets. Walk around and probe orally: 'Where on your graph does the moult happen? Why is the line flat between the steps?' WAIT TIME: 10-15 seconds before prompting further. Ask: 'What does juvenile hormone do in a caterpillar compared with ecdysone?' Cold-call 3 groups to state one piece of evidence each. Remind learners of animal welfare: any live insects are handled gently in ventilated containers and will be returned to the garden at the end of the    | Predict-Observe-Explain (observe step): learners compare their earlier predictions and Day 1 models with actual data from the exuviae, life-cycle models and graphs. Linking a flat section of the graph to a rigid exoskeleton and a vertical step to the moment of moulting helps learners see growth as controlled and intermittent, not smooth.            | Growth curve graph check: teacher checks that each group has correctly labelled axes with units, identified a step (moult) and a plateau (hard exoskeleton), and calculated one growth rate accurately using the standard formula. Oral probe indicators: learners correctly state that ecdysone triggers moulting and that high juvenile hormone gives another nymph or larval stage while low levels allow pupation or adult formation. Record groups that confuse complete and incomplete metamorphosis for reteaching. |

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| <p> <a href="#">Search ARES for readings</a><br/>Search terms: growth development animals exuvia</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>growth in insect body length and sigmoid growth in other animals such as a calf or a chicken. Each group lists on a table the evidence that explains: why the insect growth curve has steps, what happens at each moult, and how complete and incomplete metamorphosis differ. They recalculate one growth rate from the graph using the standard formula from Lesson 3 (growth rate = change in length ÷ time taken) for a step phase and a flat phase.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>lesson. Check that the growth rate calculation uses the formula from Lesson 3 with correct units (mm per day).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Organism growth and the environment</a><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/> <a href="#">Search ARES for readings</a><br/>Search terms: growth development animals exuvia</p> | <p>Each group builds a Claim-Evidence-Reasoning (CER) poster on chart paper answering the Driving Question: 'How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?' Claim: a one- or two-sentence answer. Evidence: at least three pieces drawn from the exuviae, life-cycle models, hormone information and growth curve data (including the calculated growth rates). Reasoning: a scientific explanation linking the evidence to the claim, including why controlled growth and developmental transitions help insects optimize resource use (for example, a caterpillar feeding on sukuma wiki leaves while the adult butterfly drinks nectar), reduce parent-offspring competition for food, and survive harsh environments (for example, a pupa or egg surviving the dry season in the Rift Valley). Groups then take part in a Gallery Walk: posters stay up, two members rotate to read other posters and give feedback using a peer rubric while two stay to explain.</p> | <p>Model one short CER example on the board using a different Kenyan context first (for example, why locusts moult several times): 'My claim is..., my evidence is..., my reasoning connects the evidence to the claim because...'. Give 20 minutes for poster construction. Say: 'Your reasoning must answer why, not just what.' Circulate and ask: 'How does your evidence support your claim?' WAIT TIME: 10-15 seconds. Then start the Gallery Walk: 'Rotate clockwise every 3 minutes. Leave one star (a strength) and one question on a sticky note for each poster.' Cold-call 5 learners afterwards: 'What was the strongest piece of evidence you saw on another poster?' Give groups 4 minutes to read the feedback and revise their posters.</p> | <p>Claim-Evidence-Reasoning (CER) modelling and Gallery Walk peer review: CER forces learners to connect data to scientific ideas rather than just stating facts, and the Gallery Walk exposes them to alternative explanations, so that they strengthen their reasoning and correct remaining misconceptions through discussion with peers.</p> | <p>Peer rubric (4 criteria scored 1-3): claim answers both parts of the Driving Question; at least three accurate pieces of evidence; reasoning links hormones to moulting and to metamorphosis type; explanation of survival advantage (resource use, reduced competition, harsh environments). Observable indicators: posters state that the old exoskeleton is replaced, not peeled like skin; hormones are named; a stage-by-stage contrast between complete (egg, larva, pupa, adult) and incomplete (egg, nymph, adult) metamorphosis appears. Teacher uses the rubric to spot-check the weakest two posters and offers a prompt for revision.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Organism growth and the environment</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES for readings</a><br/> Search terms: growth development animals exuvia</p> | <p>The class returns to the Driving Question Board that has been built since Lesson 1. Learners move their sticky-note questions into three columns: 'Answered (with evidence)', 'Partly answered', and 'New questions'. For each answered question, one volunteer attaches a short evidence note (a line from the CER poster). Learners add new questions that arise, for example: 'How might human pesticides disrupt these exact hormonal pathways in agricultural pests?' and 'Do other animals, such as crabs in Mombasa, moult like insects?'</p>                                                                                                                  | <p>Say: 'Let's look at our board together. Which questions can we now answer, and what is our evidence?' Allow learners to move their own notes. Cold-call 6 learners to justify why they placed a question in a column: 'What evidence tells you this question is answered?' WAIT TIME: 10-15 seconds before accepting responses. Highlight the new question on pesticides and hormones: 'This is where science goes next. Farmers in Kericho, Kisumu and the Rift Valley use pesticides on crops. What might happen if a chemical mimics juvenile hormone or blocks ecdysone?' Record the new question prominently as the bridge to future learning.</p>                                                   | <p>DQB completion with evidence-linking: tying each question to evidence shows learners how their understanding has grown across the unit and reinforces that science is an ongoing process in which answers lead to new questions.</p>                                                      | <p>Observable indicators: learners correctly sort at least 80% of questions and justify placements with specific evidence; new questions use unit vocabulary and are testable (for example, involve hormones, life stages or pesticides). Teacher checks the number of learners who contributed to the board and notes which questions remain 'partly answered' to address in revision.</p>                                                                                                                                                                                                                                            |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Organism growth and the environment</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i></p>                                                                                                                                                                                                  | <p>Each learner takes out their Day 1 model of how the insect outgrew its shell and creates a revised Final Model on a fresh sheet or on the reverse side, using labelled drawings and arrows. The final model must show: the exoskeleton and why it cannot stretch; hormonal signals (ecdysone and juvenile hormone) triggering moulting; the new, soft, larger exoskeleton hardening after the moult; the intermittent growth curve; and a comparison of complete versus incomplete metamorphosis with a Kenyan example of each. Learners then complete a 'Day 1 vs Final' comparison table with three columns: What I thought on Day 1, What I know now, and What</p> | <p>Say: 'Take your Day 1 model. With a different coloured pen, show every change that your evidence supports.' Allow 15 minutes of individual work, then 5 minutes of Think-Pair-Share to compare changes. Circulate with oral probes: 'Where does the hormone act in your model?' 'Why does a caterpillar not simply grow like a calf?' WAIT TIME: 10-15 seconds. Cold-call 4 learners to explain one change. Distribute the exit ticket: '1) In two sentences explain how an insect outgrows its exoskeleton. 2) Name one advantage of complete metamorphosis. 3) One question I still have.' Remind learners that live insects must be returned to the garden gently. Close by reading aloud the next</p> | <p>Cumulative model revision with Day 1 versus Final comparison: making learners visibly change a model, rather than starting anew, shows how evidence changes ideas and builds metacognition about their own learning. Think-Pair-Share supports peer clarification of the final model.</p> | <p>Exit ticket and model check. Success indicators: the final model shows a moulting step with hormone labels; explains that growth happens in bursts after moulting rather than smoothly; distinguishes complete from incomplete metamorphosis correctly; includes at least one survival advantage (less competition for food between larva and adult, dormancy in a pupa during dry or cold conditions, efficient use of resources). Day 1 versus Final table includes at least three specific, evidence-based changes. Learners who fail to name hormones or still say that insects 'peel' or 'melt' receive follow-up support.</p> |

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|  <b>Search ARES</b><br>for readings<br>Search terms:<br>growth<br>development<br>animals exuvia | evidence changed my thinking.<br>Finally they complete an exit ticket. | question: 'How might human pesticides disrupt these exact hormonal pathways in agricultural pests?' |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--|--|

### D. TEACHER REFLECTION

Which groups could move from describing what happens (moulting, metamorphosis) to explaining why it happens using hormones and survival advantages? Which misconceptions (insects peeling like sunburned skin, caterpillars melting and rebuilding without control) persisted in posters or models, and what evidence would address them? Did the growth rate calculations follow the standard formula from Lesson 3 consistently, and could learners interpret the steps and plateaus on the growth curve? Were the animal welfare guidelines for local insect collection followed and all live specimens returned? How effectively did the print-based data sheets support groups without internet access? What follow-up activity could connect the pesticide question to local farming practices in Kenya?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>What did I observe?</b>                   | We observed an empty exuvia with a split down its back beside a larger newly emerged grasshopper or cicada, caterpillars becoming pupae and butterflies, and insect growth curves with steps and flat sections. We also observed how our Day 1 models differed from our final models.                                                                                                                                                                                                                           |
| <b>What did I learn?</b>                     | We learned that an insect's hard exoskeleton cannot stretch, so the insect moults (ecdysis) under hormone control: ecdysone triggers moulting and juvenile hormone determines whether the next stage stays juvenile or develops toward the adult. This produces intermittent growth. Incomplete metamorphosis (egg, nymph, adult) gives small adult-like nymphs, while complete metamorphosis (egg, larva, pupa, adult) gives very different life stages that use different foods and survive harsh conditions. |
| <b>How does this explain the phenomenon?</b> | We can explain that an animal outgrows its hard shell by making a new, larger exoskeleton beneath the old one, then splitting and leaving the old exuvia while the new one is still soft and expandable. Controlled hormone-regulated growth and developmental transitions help animals optimize resource use, reduce competition between young and adults, and survive harsh environments. New question: How might human pesticides disrupt these exact hormonal pathways in agricultural pests?               |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |